

Who Gets the Better Story? An Exploration of Fairness in Explainable AI Narratives for Credit Scoring

WANNES VAN DEN BULCK, University of Antwerp, Belgium

SOFIE GOETHALS, University of Antwerp, Belgium

DAVID MARTENS, University of Antwerp, Belgium

Post-hoc explanation methods like SHAP can technically justify credit decisions but are too technical for lay people to interpret, motivating AI-generated explanatory narratives (XAIN) that use large language models (LLMs) to translate SHAP artefacts into natural language. While prior work has evaluated narrative faithfulness at the aggregate level, no study has examined whether quality differs systematically across demographic groups. We generate 204 SHAP-based narratives across six LLM providers for adversely classified instances from the German Credit dataset and break down faithfulness and feature coverage by applicant sex and age. We find substantial provider-level heterogeneity in narrative content, age- and sex-based disparities in which features are mentioned, and faithfulness differences across social groups. These exploratory findings underscore that XAIN cannot yet be deployed without structured fairness auditing and that further research is warranted.

Keywords: Fairness, Explanatory Narratives, XAI, Credit Scoring

Reference Format:

Wannes Van den Bulck, Sofie Goethals, and David Martens. 2026. Who Gets the Better Story? An Exploration of Fairness in Explainable AI Narratives for Credit Scoring. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26)*. Proceedings of Machine Learning Research, 9 pages.

1 Introduction

As AI systems are increasingly used in high-risk domains, credit scoring stands out as a case where explainability is both legally required and socially important [14]. While post-hoc explanation methods such as SHAP (SHapley Additive exPlanations), a method that quantifies how much each feature contributes to a model's prediction for a specific instance across all possible feature combinations [10], can technically justify (loan) decisions, they are often too technical for applicants to understand or use [13, 15]. This has motivated the move toward AI-generated explanatory narratives (XAIN), which use large language models (LLMs) to translate these artefacts into natural language explanatory narratives [11, 16]. Effective XAIN have a structured, story-like explanation that establishes cause-effect mechanisms between features, rather than a static description that merely lists feature importances [3, 11]. Example narratives generated for the same instance by two models (Grok-4-1-Fast and GPT-4o) are provided in Appendix A.1.

Authors' Contact Information: Wannes Van den Bulck, University of Antwerp, Antwerp, Belgium, wannes.vandenbulck@uantwerpen.be; Sofie Goethals, University of Antwerp, Antwerp, Belgium, sofie.goethals@uantwerpen.be; David Martens, University of Antwerp, Antwerp, Belgium, david.martens@uantwerpen.be.

This paper is published under the Creative Commons Attribution-NonCommercial-NoDerivs 4.0 International (CC-BY-NC-ND 4.0) license. Authors reserve their rights to disseminate the work on their personal and corporate Web sites with the appropriate attribution.

ECAF'26, September 02–September 04, 2026, Ghent, BE

© 2026 Copyright held by the owner/author(s).

A critical quality criterion for XAIN is *faithfulness*: the degree to which a narrative accurately reflects the underlying XAI artefact, and by extension the general context. This is part of the taxonomy proposed by Silvestri et al. [16] in the first systematic survey of XAIN. Faithfulness, encompassing *correctness* (factual agreement with the explainer’s attributions) and *completeness* (coverage of all relevant features), is directly measurable.

This is especially important in credit scoring, where an unfaithful narrative may mislead loan applicants about the actual drivers of an adverse decision by including mistakes or omitting important features, undermining their ability to contest or act upon it. Ichmoukhamedov et al. [7] propose reference-free automated evaluation for tabular SHAP narratives, directly comparing narrative content against the original SHAP table via rank, sign, and value agreement metrics extracted by a separate LLM.

Crucially, however, faithfulness has so far only been evaluated at the aggregate level, and no framework currently exists to assess fairness in XAIN [12, 16]. In credit scoring, sensitive attributes such as sex are legally excluded from model inputs [1], yet LLMs can be exposed to the full applicant profile when generating, e.g., narratives with personalisation objectives [7, 11]. As research consistently shows that state-of-the-art LLMs produce biased output even based on proxy variables alone [2, 4, 8], it is important to study differences in explanatory narratives between social groups.

In this extended abstract, we present ongoing work on the first empirical evaluation of between-group differences in SHAP narrative faithfulness. Using the German Credit dataset, we generate SHAP-based narratives for adversely classified instances across six LLM providers, and evaluate faithfulness using the automated metrics of Ichmoukhamedov et al. [7], extended with other attribute coverage metrics and linguistic metrics. We focus on adversely classified instances because these applicants bear the negative outcome (credit denial) and must be able to understand, contest, and act on the decision to improve their situation. Unfair narratives can cause allocational harm (disparate withholding of access to credit) and representational harm (certain groups framed in persistently more negative terms), making adverse decisions in high-risk contexts the focus for fairness auditing. We then break down all results by applicant sex and age to test whether there exist faithfulness differences in XAINs between social groups.

2 Experiments setup

We use the German Credit dataset (1,000 instances, 20 features), a widely used dataset in the fairness literature [6]. A random forest classifier was trained on a 700-instance training set, excluding the protected attributes *sex*, *age*, and *foreign_worker* (a proxy for racial/ethnic origin) as model inputs, achieving $AUC = 77.8\%$ and accuracy = 74.5% on a held-out test set of 300 instances. Of these, 34 instances were adversely classified ("denied credit"), and constitute the focus of this study.

For each of the 34 adversely classified instances, SHAP values were computed using the TreeSHAP algorithm [9] and ranked by absolute importance. Six state-of-the-art LLM providers were then prompted to generate an explanatory narrative: Claude Sonnet 4 (Anthropic), DeepSeek-Chat (DeepSeek), Gemini 3.1 Flash Lite (Google), Grok-4-1-Fast (xAI), Mistral Large (Mistral), and GPT-4o (OpenAI). These six providers span across different geographic and regulatory contexts (the United States, Europe and China). With the inclusion of DeepSeek-Chat, an open-weight model was also represented alongside the otherwise proprietary systems. All models were called at temperature $T = 0$. The prompt consists of five components, building on earlier research [7, 11, 16]. First, *dataset information* contextualises the task. Second, a *technical explanation* of SHAP values is provided. Third, the full

applicant profile is shared, listing all feature values and descriptions alongside comparisons to the distribution or average of approved applicants for each feature (except for protected attributes). With this, the applicant’s predicted probability of bad credit is given. Fourth, a *feature importance table* presents the top 3 SHAP features ranked by absolute value, along with their SHAP scores and direction of influence. Fifth, a *narrative task instruction* specifies content, constraints, and style. An overview of the XAIN generation and prompt components is given in Appendix A.2. The inclusion of protected attributes in the prompt follows from the need to pass the full applicant profile to the LLM, as coherent narrative generation requires access to the underlying feature values rather than SHAP scores alone [3, 7, 11], and can benefit personalisation objectives, explaining not only what drove the prediction, but additionally enabling cause-effect mechanisms and continuous structure that connect features to the individual’s specific situation.

To evaluate faithfulness, we adopt a slightly adapted version of the extraction-based pipeline of Ichmoukhamedov et al. [7]. Three LLMs (DeepSeek-Chat, Grok-4-1-Fast, GPT-4o) independently extract structured information from each narrative, and a majority voting function produces a final extraction per instance per generator LLM to decrease mistakes a single extractor LLM could make [7]. This extraction is then compared against a ground truth file derived directly from all features and values as passed in the generation prompt, yielding the faithfulness and other metrics described in the following section. The full generation and evaluation workflow is illustrated in Appendix A.3. As He and Martens [5] show, single-pass generation introduces faithfulness issues even for state-of-the-art LLMs. We adopt this setting deliberately to establish a baseline for between-group differences prior to any refinement.

3 Results

Six providers generated narratives for 34 adversely classified instances, yielding 204 narratives in total (126 male, 78 female; 102 young < 32 , 102 old ≥ 32). Age was split at the sample median (32 years) to give balanced subgroups. As all 34 adversely classified instances were foreign workers, the variable *foreign_worker* was necessarily excluded from analysis. Given the exploratory nature of this ongoing work and the limited sample size, all statistical tests are reported without multiple comparison correction. Feature mention disparities were assessed using Fisher’s exact test when expected cell counts fell below 5, and Chi-square tests of independence otherwise. Two-group comparisons on faithfulness and linguistic metrics used Mann-Whitney U tests; four-way subgroup comparisons used Kruskal-Wallis tests with post-hoc pairwise comparisons. All findings should be interpreted as exploratory and hypothesis-generating, pending confirmatory analysis on larger datasets.

First, there exist provider-level differences (Figure 1, left). Substantial variation exists in how providers construct narratives beyond the top-ranked SHAP features. Grok mentions the most additional features (avg 9.15), OpenAI almost none (avg 0.44). Only Gemini and Grok mention protected attributes: Gemini references age in nearly all of these cases (and sex once via a gendered pronoun), while Grok references all protected attributes.

Second, several differences emerge between the younger and older groups (Figure 1, right). Gemini constructs significantly more detailed narratives for older applicants, mentioning on average 5.88 non-protected, non-SHAP features compared to 4.94 for younger applicants ($p = 0.0265$). Further inspection shows this could be due to some features being exclusively mentioned for older applicants, including *number of credits*, *people liable*, and *credit history*. Mistral shows a significant drop in rank 3 agreement for older applicants (35.3%) versus younger (70.6%, $p = 0.0443$, Cohen’s $d = 0.733$), attributable to Mistral including time-related features such as *employment*

duration for older applicants even when these do not appear in the top-3 SHAP features, effectively confusing the recipient of the narrative. Across all providers, *people liable* is mentioned almost ten times more frequently for older applicants (19.6% vs. 2.0%, $p = 0.0001$), and *number of credits* approximately three times more often (22.5% vs. 8.8%, $p = 0.0123$).

Third, some of these differences are also to be seen between the sexes (Figure 1, right). *People liable* is mentioned in 15.9% of male narratives versus 2.6% of female narratives ($6.19\times$, $p = 0.0060$), and *number of credits* in 20.6% versus 7.7% ($2.68\times$, $p = 0.0231$). Crucially, neither feature appears in the top-3 SHAP features for any instance of either sex, so these disparities do not stem from this difference. At the provider level, Claude shows the sharpest sex differences: *number of credits* is mentioned for 38.1% of male but 0% of female narratives ($p = 0.0333$), while *housing* is mentioned for 69.2% of female but only 28.6% of male narratives ($p = 0.0494$).

Lastly, some general text metrics were calculated. Word count, sentence length, type-token ratio, Flesch-Kincaid grade, and causal connective density show no significant differences by sex or age when averaged across providers.

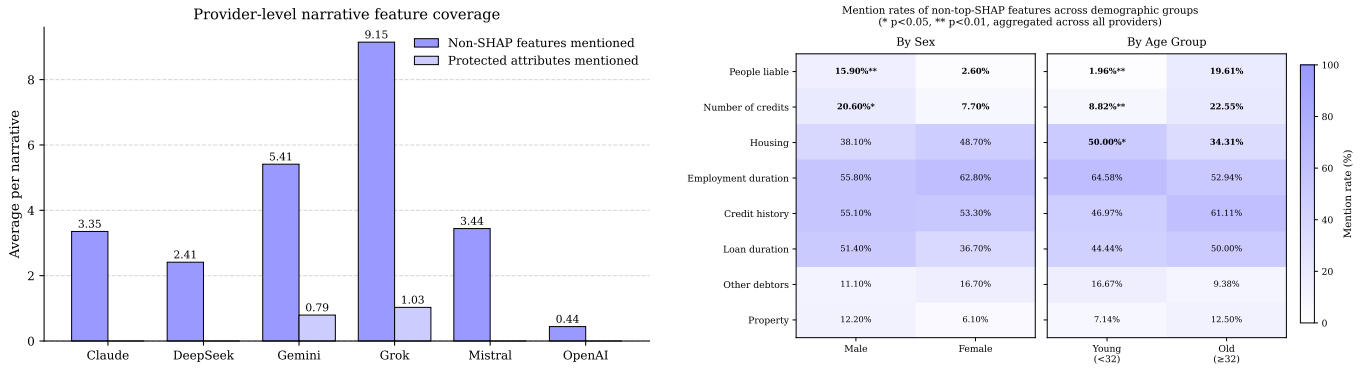


Fig. 1. Provider-level feature coverage (left) and demographic disparities in feature mention rates (right).

4 Discussion

The results reveal two distinct types of disparity in LLM-generated credit denial narratives. The first is heterogeneity in *feature coverage*: which non-SHAP features are mentioned varies substantially across providers and across demographic groups. The second is heterogeneity in *faithfulness*: rank agreement differs across social groups.

The provider-level variation in coverage is itself a fairness concern independent of demographics. An applicant whose bank uses Grok receives a qualitatively different narrative than one whose bank uses OpenAI, regardless of their credit profile. When such variation also tracks demographic groups, as with *people liable* and *number of credits* being mentioned far more often for male and older applicants, the question arises whether this reflects genuine differences in underlying feature values or LLM-internal biases. Even where demographic disparities in feature mentions are partially warranted by feature value distributions in the data, it remains problematic that providers differ: whatever a practitioner deems fair is violated by at least some LLMs, since they produce divergent results on identical inputs.

Several limitations bound these findings. First, manual human evaluation was limited: not all generated narratives were reviewed, and those that were were assessed by a single annotator. Second, the three LLMs used to extract structured information from the narratives (DeepSeek-Chat, Grok-4-1-Fast, GPT-4o) overlap with the providers used to generate them, introducing potential self-evaluation bias, though majority voting across the three extractors mitigates this. Third, the 204 narratives are not fully independent, as they derive from 34 adversely classified instances across 6 providers: comparisons pooled across the full set must therefore be interpreted with care. Combined with the absence of multiple-comparison correction, all reported p-values should be read as hypothesis-generating and exploratory rather than confirmatory.

For future work, two targeted experiments would help isolate the mechanism driving these disparities. First, injecting protected attributes explicitly into the prompt for the same instances would reveal whether their presence directly alters narrative content. Second, excluding all protected attributes from the prompt would test whether feature value co-variation alone is sufficient to produce the observed gaps. Together, these experiments would determine whether unfairness in XAINs originates from the interaction of feature values with protected attributes in the prompt, from protected attributes alone, or from feature value distributions independently.

Acknowledgments

We acknowledge the support of the Research Foundation Flanders (FWO), Grant G0G2721N.

References

- [1] 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence. <http://data.europa.eu/eli/reg/2024/1689/oj> Artificial Intelligence Act.
- [2] Xuechunzi Bai, Angelina Wang, Ilia Sucholutsky, and Thomas L. Griffiths. 2025. Explicitly unbiased large language models still form biased associations. *Proceedings of the National Academy of Sciences* 122, 8 (Feb. 2025), e2416228122. doi:10.1073/pnas.2416228122
- [3] Mateusz Cedro and David Martens. 2026. On the Importance and Evaluation of Narrativity in Natural Language AI Explanations. doi:10.48550/arXiv.2604.18311 arXiv:2604.18311.
- [4] Isabel O. Gallegos, Ryan A. Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K. Ahmed. 2024. Bias and fairness in large language models: A survey. *Computational Linguistics* 50, 3 (2024), 1097–1179. <https://direct.mit.edu/coli/article/50/3/1097/121961>
- [5] Yifan He and David Martens. 2026. An Agentic Approach to Generating XAI-Narratives. doi:10.48550/arXiv.2603.20003 arXiv:2603.20003.
- [6] Hans Hofmann. 1994. Statlog (German Credit Data). UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5NC77>.
- [7] Timour Ichmoukamedov, James Hinns, and David Martens. 2024. How good is my story? Towards quantitative metrics for evaluating LLM-generated XAI narratives. doi:10.48550/arXiv.2412.10220 arXiv:2412.10220.
- [8] Alina Leidinger and Richard Rogers. 2024. How Are LLMs Mitigating Stereotyping Harms? Learning from Search Engine Studies. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 7, 1 (Oct. 2024), 839–854. doi:10.1609/aies.v7i1.31684
- [9] Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. 2020. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* 2, 1 (Jan. 2020), 56–67. doi:10.1038/s42256-019-0138-9
- [10] Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates, Inc. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [11] David Martens, James Hinns, Camille Dams, Mark Vergouwen, and Theodoros Evgeniou. 2025. Tell me a story! Narrative-driven XAI with Large Language Models. *Decision Support Systems* 191 (April 2025), 114402. doi:10.1016/j.dss.2025.114402
- [12] Eleanor Mill, Wolfgang Garn, and Chris Turner. 2024. Real-World Efficacy of Explainable Artificial Intelligence using the SAGE Framework and Scenario-Based Design. *Applied Artificial Intelligence* 38, 1 (2024), 2430867. doi:10.1080/08839514.2024.2430867
- [13] Tim Miller, Piers Howe, and Liz Sonenberg. 2017. Explainable AI: Beware of Inmates Running the Asylum Or: How I Learnt to Stop Worrying and Love the Social and Behavioural Sciences. doi:10.48550/arXiv.1712.00547 arXiv:1712.00547.
- [14] Samuele Lo Piano. 2020. Ethical principles in machine learning and artificial intelligence: cases from the field and possible ways forward. *Humanities and Social Sciences Communications* 7, 1 (June 2020), 9. doi:10.1057/s41599-020-0501-9
- [15] Yao Rong, Tobias Leemann, Thai-Trang Nguyen, Lisa Fiedler, Peizhu Qian, Vaibhav Unhelkar, Tina Seidel, Gjergji Kasneci, and Enkelejda Kasneci. 2024. Towards Human-Centered Explainable AI: A Survey of User Studies for Model Explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 4 (April 2024), 2104–2122. doi:10.1109/TPAMI.2023.3331846
- [16] Matteo Silvestri, Vittoria Vineis, Edoardo Gabrielli, Flavio Giorgi, Fabiano Veglianti, Fabrizio Silvestri, and Gabriele Tolomei. 2025. A Survey on Explainable AI Narratives based on Large Language Models. <https://www.authorea.com/doi/full/10.36227/techrxiv.176282217.77757258>